

A

Label-Marker Agreement
LLMarkers extracted corpus

Marker Identifiability*

Exact $J = 1$ Partial $0 < J < 1$ None $J = 0$

Reported Labels	Exact	Partial	Different
Exact	45 0.0008%	2,403 0.04%	3,172 0.06%
Partial	30 0.0005%	8,228 0.15%	37,171 0.66%
Different	76 0.0014%	102,369 1.8%	443,977 97.3%

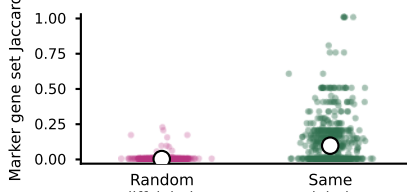
* $J = |G_1 \cap G_2| / |G_1 \cup G_2|$

- same profile
- same label, different markers
- alias / naming variation
- context clarifies identifiability
- different profile

B

Inconsistent Pairwise Labels
LLMarkers extracted corpus

Same-label marker sharing

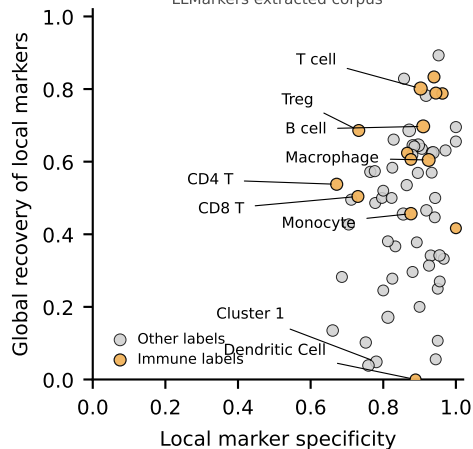


Different labels, high marker overlap

Reported labels	J	Shared	Pairs
Naive T / TCM	1.00	4/4	2
B cell / B cells	1.00	3/3	7
Monocyte / Monocytes	1.00	3/3	6
Oligo. / Oligo. cells	1.00	3/3	4
Epi. / Epithelial	1.00	3/3	2
RBC / Erythroid	1.00	3/3	2

C

Globally Identifiable Markers
LLMarkers extracted corpus



D

Label-Marker Agreement
CAP human profiles, cross-project

Marker Identifiability*

Exact $J = 1$ Partial $0 < J < 1$ None $J = 0$

Reported Labels	Exact	Partial	Different
Exact	2 0.0003%	208 0.03%	97 0.01%
Partial	0 0.0000%	2,998 0.39%	6,711 0.88%
Different	0 0.0000%	13,598 1.8%	736,533 96.9%

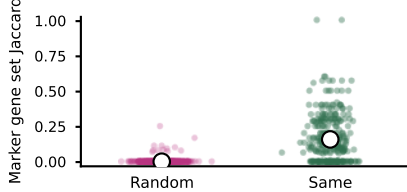
* $J = |G_1 \cap G_2| / |G_1 \cup G_2|$

- same profile
- same label, different markers
- alias / naming variation
- context clarifies identifiability
- different profile

E

Inconsistent Pairwise Labels
CAP human profiles, cross-project

Same-label marker sharing



Different labels, high marker overlap

Reported labels	J	Shared	Pairs
cDC2 / DC2	0.75	3/4	3
Mast Cell / Mast Cells	0.75	3/4	2
BEST4 epithelial / BEST4 colonocytes	0.75	3/4	2
Interstitial Mph / Pvms	0.67	4/6	5
cDC2 / cDC2B	0.60	3/5	15
cDC2 / type 2 cDCs	0.60	3/5	4

F

Globally Identifiable Markers
CAP ontology terms, cross-project

